

Semantic Identity Compression: Exact Zero-Error Laws, Rate-Distortion, and Neurosymbolic Necessity

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Abstract

We prove that no fixed semantic representation, however sophisticated, can achieve zero-error identity recovery without explicit auxiliary information. Any non-injective representation creates an irreducible identity debt: the worst-case auxiliary description requires at least $\log_2 A_\pi$ bits, where A_π is the maximum collision-fiber size. If those bits are withheld, the distortion floor on the worst fiber is at least $1 - 1/A_\pi$.

This paper shows that a single structural object, the collision fiber geometry, governs all operational costs for exact identity: the bit cost (fixed-length and adaptive), the distortion floor, and the query complexity for operational disambiguation. On the query side, the paper proves that complete derivability-aware query systems reduce to orthogonal minimal cores, so the query cost is structural rather than presentation-dependent. These are alternate currencies for the same residual ambiguity; none yields a free way to restore identity.

All main results are machine-checked in Lean 4. The formalization both verifies correctness and demonstrates that the fiber-geometry framework is conceptually clean enough for full mechanization.

Keywords: semantics-aware compression, zero-error coding, neurosymbolic systems, learned representations, side information

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I. INTRODUCTION

A. Problem Statement

We prove a simple fact: no learned semantic representation can, by itself, guarantee zero-error identity recovery in an open world unless it is injective. When distinct identities share the same representation value, the representation leaves behind an irreducible identity debt. That debt must be paid in one of several currencies: auxiliary bits, interactive queries, or accepted distortion. There is no third free option.

This paper isolates the structural object that makes that statement exact: the collision fiber geometry induced by a fixed representation map π . Once that object is explicit, a family of exact finite theorems follows. The same fiber geometry yields the fixed-length converse $L \geq \log_2 A_\pi$, the fiberwise adaptive budget $\lceil \log_2 |\pi^{-1}(u)| \rceil$, an explicit deterministic distortion floor when the identity bits are withheld, and a query theorem showing that every complete derivability-aware query system reduces to an orthogonal minimal core on which matroid structure emerges.

The systems consequence is immediate. Symbolic handles, identifiers, keys, pointers, registry entries, and retrieval indices are not merely convenient engineering artifacts. They are the standard mechanisms by which real systems pay the identity debt left by non-injective semantic abstraction.

Roadmap. Section II defines the information barrier and the collision fibers. Section III gives the exact coding laws for fixed-length and adaptive side information. Section IV shows that the query currency canonically reduces to an orthogonal minimal core and that all canonical minimal distinguishing families are equicardinal. Section V derives the deterministic distortion floor and its entropy-sensitive companions. Section VI explains the neurosymbolic and systems consequences. Section VII studies open-world fragility. Section VIII closes the paper.

B. Why This Is a Compression Problem

Let C be a finite class label and let $U = \pi(C)$ be the representation available to the decoder. Exact identity recovery asks how much additional description is needed to recover C from U .

Once one writes the problem this way, the central combinatorial quantity is immediate:

$$A_\pi := \max_u |\pi^{-1}(u)|,$$

the size of the largest collision fiber. The worst-case fixed-length law is then exactly the minimum number of extra bits needed to separate that worst fiber:

$$L \geq \log_2 A_\pi.$$

This viewpoint also yields a finer adaptive law. If the auxiliary description length may depend on the realized representation value u , then the optimal pointwise budget is controlled by the size of the individual fiber $\pi^{-1}(u)$. The same geometry also determines what happens when the budget is subcritical: if one refuses to pay the necessary identity bits, the distortion floor is explicit. In this sense, worst-case bits, adaptive bits, query cost, and deterministic distortion are all different views of the same residual ambiguity object.

All results are stated relative to one admissibility contract: the decoder may use only the declared representation value and any auxiliary description whose length is charged to the budget. No hidden registry, preprocessing table, or external side channel may contribute uncharged distinguishing information.

C. Relation to Prior Work

The closest classical objects are zero-error coding with confusability constraints [1], [2], source coding with side information, and more recent task-oriented or semantics-aware communication models [3]. Our setting isolates a different question: given a fixed deterministic representation already available at the decoder, what residual coding burden does it leave behind for exact identity recovery? That question admits exact finite answers.

The confusability graph in our setting is highly special: classes sharing one representation value form a clique, so the graph is a disjoint union of representation-fiber cliques. This structural collapse explains why the paper obtains exact closed forms that the general graph-entropy problem does not. The modern interpretation is neurosymbolic necessity: if π is a learned

embedding, then collision fibers measure the irreducible identity ambiguity left by semantic compression. Injective representations need no auxiliary metadata; non-injective representations impose an exact-recovery cost that no downstream reasoning layer can remove on its own [4], [5].

D. Contributions

Main contribution. The collision fiber geometry induced by a fixed representation π is the single structural object that governs exact identity-recovery costs. From this object flow several direct facets:

- 1) Exact coding laws: a tight fixed-length converse $L \geq \log_2 A_\pi$, an exact finite-block scaling law, and a pointwise optimal adaptive bit budget $\lceil \log_2 |\pi^{-1}(u)| \rceil$ per fiber.
- 2) Deterministic distortion currency: in the uniform single-block corner the optimal distortion is $D^*(L) = \max(0, 1 - 2^L/a)$, giving an explicit error floor when identity bits are withheld.
- 3) Canonical query currency: every complete derivability-aware query family reduces to an orthogonal semantically minimal core, so non-orthogonal families are redundant overpresentations rather than a rival regime. On that canonical core, minimal distinguishing families form the bases of a matroid, and the query cost is an exact structural invariant.
- 4) Systems payoff with mechanized certainty: symbolic handles are the canonical mechanism for paying the identity debt in real systems, and the main theorem chain is machine-checked in Lean 4.

a) Machine-checked proofs.: All principal theorems in this paper have been formalized and verified in Lean 4. The mechanization functions as an integrity gate: informal arguments were admitted into the paper only after the corresponding formal proofs compiled without `sorry`.

E. Worked Example

We use one small example throughout the paper. Five classes have profiles

$$A = 000, \quad B = 000, \quad C = 100, \quad D = 110, \quad E = 111.$$

Then A and B form a collision fiber of size two, so the exact worst-case fixed-length budget is one bit. The adaptive picture is equally simple: the fiber over profile 000 needs one auxiliary bit, while the singleton fibers need none. In the learned-representation reading, this is the exact price of one residual collision in the representation.

F. Paper Organization

The paper has one theoremic arc and several direct payoffs. Sections II and III establish the exact coding backbone. Section IV shows that the query-side cost is canonical once redundant observables are collapsed. Section V converts the same ambiguity into a deterministic distortion floor and entropy-sensitive converses. Section VI explains why symbolic identity layers are therefore necessary in semantics-aware systems. Section VII studies open-world fragility. Section VIII then closes the paper.

II. MODEL AND BASIC INFORMATION BARRIER

A. The Abstract Coding Framework

The paper studies a finite class space together with a fixed representation map. The decoder observes the representation value, and the encoder may additionally send auxiliary description to resolve whatever ambiguity the representation leaves behind. The coding laws depend only on this representation map and its fibers, not on the mechanism that produced it.

Definition II.1 (Coding model induced by a representation). Let C be a finite class variable, let $U = \pi(C)$ be the representation available to the decoder, and let M be any auxiliary description transmitted by the encoder. The main theorem arc studies the minimum length of M needed for exact recovery of C from (U, M) , in both fixed-length and representation-adaptive regimes.

Definition II.2 (Representation fiber). For a representation value u , define the corresponding fiber by

$$\pi^{-1}(u) = \{c : \pi(c) = u\}.$$

The paper's central object is the geometry of these fibers.

Remark II.3 (Where the representation may come from). No further structure is assumed. The map π may arise from a learned encoder, a database schema projection, a feature-extraction pipeline, or a hand-designed attribute family. The theoremic contribution is generic in the representation map once its fibers are fixed.

Remark II.4 (Entity and random-variable notation). The coding statements are written in terms of a class variable C and representation $U = \pi(C)$. When convenient, we also use an underlying entity variable V together with a class assignment map $C(V)$. Thus $U = \pi(V)$ is the representation of the realized entity, and the induced class label is $C(V)$. The two viewpoints are interchangeable; the coding arc should be read through the pair (C, U) .

Definition II.5 (Admissibility contract). All impossibility and optimality statements in this paper quantify over schemes whose evidence consists only of the declared representation value together with any explicit auxiliary description charged to the side-information budget. In particular, no hidden registry, reflection oracle, whole-universe preprocessing table, or amortized cross-instance cache may supply uncharged distinguishing information.

B. Representation-Induced Information Barrier

The basic obstruction is immediate: equal representations give the decoder the same side information. The next theorem records that invariant in generic form.

Definition II.6 (Representation-computable property). A property $P : \mathcal{C} \rightarrow Y$ is *representation-computable* if there exists a function $f : \mathcal{U} \rightarrow Y$ such that

$$P(c) = f(\pi(c)) \quad \text{for all } c \in \mathcal{C}.$$

Equivalently, P depends only on the representation value.

Theorem II.7 (Information barrier). *Let $P : \mathcal{C} \rightarrow Y$ be any function. If P is representation-computable, then P is constant on every representation fiber:*

$$\pi(c_1) = \pi(c_2) \implies P(c_1) = P(c_2).$$

Equivalently, no observer restricted to the representation alone can compute a property that varies within one fiber.

Proof. Suppose P is representation-computable via f , so $P(c) = f(\pi(c))$ for all c . If $\pi(c_1) = \pi(c_2)$, then

$$P(c_1) = f(\pi(c_1)) = f(\pi(c_2)) = P(c_2).$$

■

Remark II.8 (Role of the barrier). The barrier is informational rather than computational: once the representation is identical, unlimited time or memory does not help. We state it explicitly because the later converse, adaptive coding, query, and robustness results all reuse the same invariance premise.

Corollary II.9 (Class identity is not representation-computable). *If there exist classes c_1, c_2 with $\pi(c_1) = \pi(c_2)$ but $c_1 \neq c_2$, then class identity is not representation-computable.*

Proof. Direct application of Theorem II.7 to the identity map on classes. ■

Theorem II.10 (Observable-state sufficiency). *Let $\alpha : \mathcal{C} \rightarrow \mathcal{A}$ be any declared observable state such that every admissible primitive observation factors through α . Then every in-scope semantic property factors through α : there exists $\tilde{P}$ with*

$$P(c) = \tilde{P}(\alpha(c)) \quad \text{for all } c \in \mathcal{C}.$$

Proof. If every admissible primitive observation factors through α , then every admissible transcript factors through α , and so does any output computed from that transcript. ■

Corollary II.11 (Observable-state insufficiency). *If $\alpha(c_1) = \alpha(c_2)$ but $P(c_1) \neq P(c_2)$, then no admissible strategy can compute P with zero error without adding new information beyond the declared observable state.*

Remark II.12 (Interpretation). The sufficiency and insufficiency statements are the helper-view version of the information barrier: if a task varies inside one declared state fiber, that state is

not sufficient and must be refined by additional information.

III. CODING CONSEQUENCES: BIT CURRENCY

A. Fixed-Length Zero-Error Recovery

We now state the zero-error coding problem in generic representation-map form. Let C be a finite class variable and let $U = \pi(C)$ be the representation observed by the decoder. If the representation is non-injective on classes, then the decoder cannot determine C from U alone; the unresolved ambiguity is exactly the ambiguity inside the fibers of π .

Definition III.1 (Observation channel). Let C be a random class label, let $U = \pi(C)$ be its representation, and let the decoder observe U together with any charged auxiliary description supplied by the encoder.

Theorem III.2 (Zero auxiliary description iff injective representation). *Let $\mathcal{C}$ be a finite class space and let $\pi : \mathcal{C} \rightarrow \mathcal{U}$ be a representation map. Zero-error recovery of C from $U = \pi(C)$ with no auxiliary description is achievable if and only if π is injective on classes.*

(L: ACS8)

Proof. Achievability: If π is injective on classes, then observing $\pi(c)$ determines c uniquely. The decoder simply inverts the class-to-representation map.

Converse: Suppose two distinct classes $c_1 \neq c_2$ share the same representation value: $\pi(c_1) = \pi(c_2)$. Then any decoder $g(U)$ outputs the same class label on both inputs, so it cannot be correct for both. Hence zero-error recovery without auxiliary description is impossible. ■

In the finite explicit setting, the same condition can be read directly from the representation channel: one-symbol zero-error feasibility is equivalent to injectivity (GPH27). Interpreting π as a learned embedding, this is the exact threshold for zero-error class recovery from the representation alone.

Remark III.3 (Information-theoretic corollary). Under any distribution with positive mass on both colliding classes, $I(C; U) < H(C)$. This is an average-case consequence of the deterministic barrier above.

(L: ACS8)

Remark III.4 (Relation to Ahlswede-Dueck). In the identification paradigm of [6], the decoder asks “is the message m ?” rather than “what is the message?” This yields double-exponential codebook sizes. Our setting is different: we require zero-error identification of the *class*, not hypothesis testing. The one-shot zero-error identification feasibility threshold is binary rather than asymptotic.

(L: ACS8)

B. Fixed-Length Auxiliary Description

We now show that augmenting the representation with an auxiliary codeword restores exact recovery.

Definition III.5 (Auxiliary description). An *auxiliary description* is a value $M(c)$ associated with each class $c \in \mathcal{C}$ and transmitted in addition to the representation value $\pi(c)$. In the fixed-length setting, the auxiliary description alphabet has size 2^L .

(L: ACS7)

Theorem III.6 (Fixed-length sufficiency). *If $|\mathcal{C}| = k$, then an auxiliary description of length $L \geq \lceil \log_2 k \rceil$ bits suffices for zero-error identification, regardless of whether π is class-injective.*

(L: ACS7)

Proof. A class-injective auxiliary description assigns a unique identifier to each class. Reading that description together with the representation determines the class exactly. ■

C. Ambiguity Converse

The next theorem is the paper’s main zero-error compression statement. It converts representation collision multiplicity directly into a necessary fixed-length side-information budget.

In graph terms, the fixed-representation setting is a structured zero-error special case: confusable classes are exactly those lying in the same representation fiber, so the confusability

graph is a disjoint union of cliques. This special structure yields exact finite laws for the residual coding burden left by a representation.

Definition III.7 (Collision multiplicity). Let $\mathcal{C}$ be a finite class space and let $\pi : \mathcal{C} \rightarrow \mathcal{U}$ be a representation map. Define

$$A_\pi := \max_{u \in \text{Im}(\pi)} |\{c \in \mathcal{C} : \pi(c) = u\}|.$$

Thus A_π is the size of the largest class-collision block under the representation.

(L: [LWDC1](#), [LWDC3](#))

Remark III.8 (Computability under finite precision). Every implementable representation operates on finite-precision values. The representation alphabet is therefore finite, and A_π is computable by fiber enumeration. If the deployed representation uses M output values, then the pigeonhole principle yields the lower bound $\lceil |\mathcal{C}|/M \rceil \leq A_\pi$. Conversely, to guarantee $A_\pi \leq k$ on an entity set of size n , at least $\lceil n/k \rceil$ distinct representation values are necessary. The coding theorems identify the operational meaning of A_π once the representation is fixed at its deployed precision.

(L: [GPH44-45](#), [GPH48](#), [GPH50](#))

If one observation channel factors through a richer one, collision multiplicity cannot increase ([GPH28](#)). Thus adding observable structure can only lower the corresponding zero-error tag bound. Under the same representation reading, enriching the embedding cannot worsen zero-error separability.

Corollary III.9 (Coarsening a bottleneck cannot reduce residual coding cost). *Let π_{fine} and π_{coarse} be two representations on the same class domain, and assume*

$$\pi_{\text{coarse}} = f \circ \pi_{\text{fine}}$$

for some map f . Then

$$A_{\pi_{\text{fine}}} \leq A_{\pi_{\text{coarse}}}, \quad \log_2 A_{\pi_{\text{fine}}} \leq \log_2 A_{\pi_{\text{coarse}}}.$$

Moreover, for each fine representation value u ,

$$\lceil \log_2 |\pi_{\text{fine}}^{-1}(u)| \rceil \leq \lceil \log_2 |\pi_{\text{coarse}}^{-1}(f(u))| \rceil.$$

(L: [GPH28](#), [GPH37](#))

Proof. The first inequality is the monotonicity of collision multiplicity under factorization. Taking binary logarithms gives the worst-case fixed-length comparison. The pointwise adaptive comparison is the same monotonicity statement applied fiberwise before taking ceiling logarithms.

■

Definition III.10 (Maximal-Barrier Regime). The domain is *maximal-barrier* if $A_\pi = k$, i.e., all classes collide under the observation map.

(L: [LWDC3](#))

Theorem III.11 (Converse). *For any classification domain, any fixed-length scheme achieving $D = 0$ requires*

$$2^L \geq A_\pi \quad \text{equivalently} \quad L \geq \log_2 A_\pi,$$

where A_π is the collision multiplicity.

(L: [LWDC1](#))

Proof. Fix a collision block $G \subseteq \mathcal{C}$ with $|G| = A_\pi$ and identical representation profile. For classes in G , the decoder's side information is identical, so zero-error decoding must separate those classes using auxiliary description outcomes. With L bits there are at most 2^L outcomes, hence $2^L \geq |G| = A_\pi$. ■

Remark III.12 (Coding interpretation). The converse is a lossless coding law for semantic identity. The relevant confusable alphabet is the largest representation-collision block, and zero-error separation of that block requires exactly enough auxiliary outcomes to name its members. The theorem therefore has the same fixed-length form as a classical zero-error side-information

budget, but the governing alphabet size is induced by the representation rather than by the ambient class set alone.

Corollary III.13 (Maximal-Barrier Converse). *If the domain is maximal-barrier ($A_\pi = k$), any zero-error scheme satisfies $L \geq \log_2 k$.*

(L: [LWDC3](#))

In the running example, the collision block $\{A, B\}$ shows that $A_\pi = 2$, so Theorem [III.11](#) already forces $L \geq 1$ bit of auxiliary description for zero-error identification.

D. Exact Finite-Block Law

The ambiguity converse extends exactly to blocks, so the main compression law is not merely one-shot. The next theorem family gives exact finite-block scaling rather than an asymptotic approximation.

Theorem III.14 (Exact one-shot feasibility threshold). *For an alphabet of n auxiliary descriptions, zero-error class recovery is feasible if and only if*

$$A_\pi \leq n.$$

(L: [GPH13](#))

Proof. By definition, two classes are adjacent in the confusability graph exactly when they have the same representation profile and are distinct. Hence each nonempty profile fiber is a clique, and there are no edges between different fibers. The confusability graph is therefore a disjoint union of profile-fiber cliques. Zero-error auxiliary description is equivalent to a proper coloring of this graph, so an alphabet of size n is feasible exactly when each fiber can be colored injectively, i.e., exactly when every profile fiber has size at most n . Taking the maximum over fibers gives the criterion $A_\pi \leq n$. ■

Theorem III.15 (Exact block feasibility law). *For block length $t \geq 1$ with coordinatewise observation on $\mathcal{C}^t$, zero-error auxiliary description with alphabet size n_t is feasible if and only if*

$$A_\pi^t \leq n_t.$$

Equivalently, the minimal feasible block alphabet is exactly A_π^t .

(L: [GPH16-17](#), [GPH20](#))

Corollary III.16 (Linear log-bit scaling). *Let $L_t^* := \log_2(A_\pi^t)$ be the minimum block auxiliary-description budget in bits. Then*

$$L_t^* = t \log_2 A_\pi, \quad \frac{L_t^*}{t} = \log_2 A_\pi.$$

So per-entity tag rate is exactly stable across block length.

(L: [GPH18-19](#), [GPH24](#))

Remark III.17 (Why this block model). The block construction repeats the same semantic identification task across t entities while keeping the representation map fixed and applying it coordinatewise. It is therefore the direct block analogue of repeated one-shot classification in this model: the collision structure multiplies across coordinates, and the required zero-error side-information budget scales accordingly.

E. Why the General Zero-Error Theory Simplifies Here

The confusability graph induced by a fixed deterministic representation has special structure: two classes are adjacent exactly when they share the same representation value. The next observations explain why several general zero-error graph quantities collapse in this setting.

Proposition III.18 (Transitivity of representation confusability). *Let $\pi : \mathcal{C} \rightarrow \mathcal{U}$. Define $c_1 \sim_\pi c_2$ iff $\pi(c_1) = \pi(c_2)$ and $c_1 \neq c_2$. Then $\sim_\pi$ is transitive on distinct classes inside a fiber.*

Proof. If $\pi(c_1) = \pi(c_2)$ and $\pi(c_2) = \pi(c_3)$, then $\pi(c_1) = \pi(c_3)$ by equality. Thus any two distinct classes linked through one fiber are directly confusable as well. ■

Proposition III.19 (Cluster-graph structure). *The confusability graph G_π , with edges between distinct classes sharing the same representation, is a disjoint union of cliques. Each connected component is exactly one representation fiber.*

(L: [GPH52-53](#))

Proof. Each fiber is a clique because all of its classes share the same representation value. No edge connects distinct fibers by construction. Hence each connected component is exactly one fiber, and the graph is a disjoint union of cliques. ■

Corollary III.20 (Capacity collapse for representation-induced graphs). *For a representation-induced cluster graph, the standard one-shot zero-error parameters collapse to fiber counts, so the fixed-length auxiliary-description law is governed exactly by the largest fiber.*

Proof. In a disjoint union of cliques, proper coloring, clique size, and independence counting reduce to choosing or naming one element per component. In this setting, the general graph inequalities collapse to the same fiber-counting law. ■

In the running example, $A_\pi = 2$, so the exact block law gives $L_t^\star = t$ bits.

F. Representation-Adaptive Side Information

The fixed-length theorem pays for the worst collision fiber through A_π . A finer coding law allows the auxiliary description length to depend on the observed representation value u . Then the relevant budget is not the maximum fiber size but the size of the individual fiber $\pi^{-1}(u)$. The same residual ambiguity object is now resolved fiber by fiber and then averaged under the induced law of U .

Definition III.21 (Representation-adaptive auxiliary description). *An adaptive auxiliary scheme assigns to each representation value u a binary description budget $\ell(u)$ and a decoder that, given*

u together with a binary string of length at most $\ell(u)$, recovers the class exactly whenever $\pi(C) = u$.

Theorem III.22 (Pointwise optimal adaptive bit budget). *For a representation fiber of size $|\pi^{-1}(u)|$, zero-error recovery under a binary auxiliary alphabet is feasible if and only if*

$$|\pi^{-1}(u)| \leq 2^{\ell(u)}.$$

Consequently, the pointwise optimal number of auxiliary bits on fiber u is

$$\ell^*(u) = \lceil \log_2 |\pi^{-1}(u)| \rceil.$$

(L: [GPH32](#))

Proof. Once u is fixed, the decoder only needs to distinguish classes inside the fiber $\pi^{-1}(u)$. With $\ell(u)$ binary bits there are at most $2^{\ell(u)}$ auxiliary outcomes, so exact recovery is possible exactly when that many outcomes suffice to name the fiber elements injectively. The minimal feasible budget is therefore the least integer whose power of two dominates the fiber size, namely $\lceil \log_2 |\pi^{-1}(u)| \rceil$. ■

Definition III.23 (Expected adaptive description length). Let $U = \pi(C)$ be the induced representation random variable under a source law on classes. For an adaptive budget ℓ , define its expected auxiliary length by

$$\mathbb{E}[\ell(U)] = \sum_u P_U(u) \ell(u).$$

Theorem III.24 (Adaptive expected-rate lower bound). *Any feasible adaptive auxiliary scheme satisfies*

$$\mathbb{E}[\ell(U)] \geq \mathbb{E}[\lceil \log_2 |\pi^{-1}(U)| \rceil].$$

Moreover, the pointwise optimal budget from Theorem [III.22](#) attains equality.

(L: [GPH32-33](#))

Proof. By Theorem III.22, feasibility on fiber u requires

$$\ell(u) \geq \lceil \log_2 |\pi^{-1}(u)| \rceil$$

for every representation value u . Multiplying pointwise by $P_U(u)$ and summing over u gives the lower bound on expected length. Equality is achieved by choosing the pointwise optimal budget on every fiber. ■

Corollary III.25 (Conditional-entropy lower bound). *Let C be drawn from any source distribution and let $U = \pi(C)$. Then any feasible adaptive auxiliary scheme satisfies*

$$\mathbb{E}[\ell(U)] \geq \frac{H(C | U)}{\log 2},$$

where $H(C | U)$ is conditional Shannon entropy in natural-log units.

(L: GPH34)

Proof. For each representation value u , the conditional law of C given $U = u$ is supported on the fiber $\pi^{-1}(u)$, so its entropy is at most $\log |\pi^{-1}(u)|$. Averaging over u gives

$$H(C | U) \leq \mathbb{E}[\log |\pi^{-1}(U)|].$$

Since $\log |\pi^{-1}(u)| \leq \lceil \log_2 |\pi^{-1}(u)| \rceil \log 2$ fiberwise, Theorem III.24 yields the stated lower bound. ■

Theorem III.26 (Fiberwise prefix-coding upper bound). *Assume standard prefix-coding achievability on each finite fiber. Then there exists an adaptive zero-error auxiliary code satisfying*

$$\mathbb{E}[\ell(U)] \leq \frac{H(C | U)}{\log 2} + 1.$$

(L: GPH35)

Proof. Because the decoder already knows $U = u$, prefix-freeness is required only within each fiber $\pi^{-1}(u)$. Applying a standard prefix-coding theorem separately to the conditional source on

each nonempty fiber gives an expected conditional code length at most $H(C \mid U = u)/\log 2 + 1$. Averaging over u yields the stated bound. ■

Corollary III.27 (Zero-error conditional-entropy sandwich). *The optimal expected zero-error auxiliary rate lies between*

$$\frac{H(C \mid U)}{\log 2} \quad \text{and} \quad \frac{H(C \mid U)}{\log 2} + 1.$$

(L: ZEC1)

Proof. The lower bound is Theorem III.24, and the upper bound is Theorem III.26. The two statements together give the claimed sandwich. ■

Corollary III.28 (Active binary clarification). *If, after observing U , the decoder is allowed to ask adaptive binary clarification questions whose leaves identify the class exactly, then the minimum expected number of clarification questions is bounded between*

$$\frac{H(C \mid U)}{\log 2} \quad \text{and} \quad \frac{H(C \mid U)}{\log 2} + 1.$$

(L: GPH34-35)

Proof. Each binary clarification protocol induces a binary code over the conditional source on each fiber, and conversely any fiberwise binary prefix code can be implemented as an adaptive yes/no decision tree. The lower and upper bounds therefore follow from the preceding conditional-entropy comparison and fiberwise prefix-coding theorem. ■

Corollary III.29 (Representation-respecting noisy transcripts cannot beat the fiber law). *Let T be any repeated or noisy observation transcript whose realized value factors through the fixed representation, that is, $T = h(U)$ for some map h . Then recovering C from T cannot require less worst-case or pointwise adaptive side information than recovering C from U itself.*

(L: GPH28, GPH37)

Proof. If the entire transcript factors through U , then it is a coarsening of the original representation. Corollary III.9 applies directly: coarsening cannot reduce the largest collision fiber and cannot reduce the pointwise adaptive fiber budget. Thus repeated or noisy observations that never break the original fibers cannot remove the residual zero-error coding burden. ■

Remark III.30 (Bit currency at fiber resolution). The fixed-length law pays for the worst collision fiber; the adaptive law pays for each fiber separately and averages under the source-induced law of U . These are two resolutions of the same object: the cost of naming the classes that remain collapsed inside each representation fiber.

Remark III.31 (Entropy comparison). The conditional-entropy lower and upper bounds place the adaptive theorem inside classical source-coding language. Zero-error adaptive side information is sandwiched around $H(C | U)$ up to the standard one-bit prefix-coding overhead, but the exact zero-error law remains fiberwise and combinatorial rather than purely entropic.

IV. QUERY CURRENCY AND CANONICAL ORTHOGONAL CORE

A. The Query Currency

Bits are one way to pay the identity debt left by a non-injective representation. Another is to interrogate primitive observables until the class is determined. This section studies that query currency. The same residual ambiguity that forces auxiliary bits can also be paid for through interactive disambiguation.

Definition IV.1 (Query family). Let $\mathcal{Q}$ be the family of primitive queries available to an observer. For a classification system with attribute set $\mathcal{I}$, we write $\mathcal{Q} = \{q_I : I \in \mathcal{I}\}$ where $q_I(v) = 1$ iff v satisfies attribute I .

Definition IV.2 (Distinguishing family). A subset $S \subseteq \mathcal{Q}$ is *distinguishing* if, for all values v, w with $\text{class}(v) \neq \text{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$.

Definition IV.3 (Minimum distinguishing number). The *minimum distinguishing number* is

$$d := \min\{|S| : S \subseteq \mathcal{Q} \text{ is distinguishing}\}.$$

Equivalently, d is the smallest number of primitive queries that suffices for zero-error separation.

Remark IV.4 (Inclusion-minimal versus minimum-cardinality). These notions differ: an inclusion-minimal distinguishing family may be larger than a minimum-cardinality one. The operational lower bound below depends only on the quantity d .

In the running example from Section III, $\{q_1, q_2, q_3\}$ is distinguishing and no pair of primitive queries is distinguishing, so the minimum distinguishing number is $d = 3$.

Theorem IV.5 (Attribute-only lower bound). *For attribute-only observers, zero-error class identity checking requires*

$$W(\text{class-identity}) = \Omega(d)$$

in the worst case, where d is the minimum distinguishing number.

(L: ACSI)

Proof. Assume a zero-error attribute-only procedure halts after fewer than d queries on every execution path. Fix one execution path and let $Q \subseteq \mathcal{I}$ be the set of queried attributes on that path, so $|Q| < d$. By definition of d , no set of size less than d is distinguishing. Hence there exist values v, w from different classes with identical answers on all attributes in Q . An adversary can answer the procedure's queries consistently with both v and w , so the transcript and output coincide on two different classes, contradicting zero-error identification. ■

Corollary IV.6 (Lower bound on attribute-only witness cost). *For any attribute-only observer,*
 $W(\text{class-identity}) \geq d$.

(L: ACSI)

This is the operational query law in full generality. It says how much interrogation is needed when the representation leaves ambiguity unresolved and no explicit identity bits are transmitted.

B. Canonical Reduction to the Orthogonal Core

The right structural invariant is not the size of an arbitrary raw query family, because raw families may contain semantically derivable redundancy. The source Lean formalization proves a stronger and cleaner fact: every semantically complete derivability-aware query family reduces to an orthogonal semantically minimal core. Non-orthogonal families are therefore redundant overpresentations, not a rival regime.

Definition IV.7 (Derivability-aware axis presentation). A *derivability-aware axis presentation* specifies primitive observables together with a derivability relation recording when one observable can be recovered from another. A family is *orthogonal* if no primitive observable is derivable from a distinct primitive observable in the family.

Proposition IV.8 (Non-orthogonal complete families have removable redundancy). *If a semantically complete derivability-aware query family is non-orthogonal, then some primitive observable can be removed without destroying semantic completeness.*

(L: L6)

Proof. If the family is non-orthogonal, some observable a is derivable from a distinct observable b already in the family. Any query requirement satisfied via a can therefore be satisfied via b instead. Erasing a preserves semantic completeness, so a was redundant. ■

Proposition IV.9 (Existence of a minimal complete core). *Every semantically complete derivability-aware query family contains a semantically minimal complete subfamily.*

(L: L7)

Proof. The family is finite. If it is not already semantically minimal, Proposition IV.8 removes a redundant observable while preserving completeness. Repeating this elimination process must terminate, producing a semantically minimal complete subfamily. ■

Proposition IV.10 (Canonical orthogonal core). *Every semantically complete derivability-aware query family contains an orthogonal semantically minimal complete subfamily.*

(L: L8)

Proof. Choose a semantically minimal complete subfamily using Proposition IV.9. If that subfamily were still non-orthogonal, Proposition IV.8 would remove another observable while preserving completeness, contradicting minimality. Hence the minimal core is orthogonal. ■

These three propositions identify the canonical regime of the query problem. One starts from an arbitrary complete family, removes derivable redundancy until no further reduction is possible, and arrives at an orthogonal minimal core. The clean query invariant lives on that core.

Theorem IV.11 (Matroid of canonical distinguishing families). *For the canonical orthogonal core of a complete derivability-aware query system, the semantically minimal distinguishing families are the bases of a matroid. Consequently, all such families have equal cardinality.*

(L: L1, L4-5, L8)

Proof. By Proposition IV.10, every complete derivability-aware query system contains an orthogonal semantically minimal core. On an orthogonal core, exchange holds (L4, L5). Standard matroid theory then implies that the minimal complete families are exactly the bases of a matroid and therefore all have the same cardinality (L1). ■

Remark IV.12 (Why the unrestricted counterexample does not undercut the theorem). Arbitrary overcomplete binary query families need not themselves be matroidal; a finite counterexample exists (GPH31). That does not refute the canonical query invariant. It shows only that redundant raw presentations can obscure the primitive structure. After derivable redundancy is stripped away, the orthogonal minimal core is matroidal.

C. Witness Cost Comparison

Remark IV.13 (Query cost as alternative currency). The bit law measures the residual description cost of restoring identity. The query law measures the residual interrogation cost. In the raw attribute model, that cost is lower-bounded by d . In the canonical derivability-aware presentation, the same cost is certified by matroid basis cardinality. In both readings, the ambiguity being paid for is the same.

Observer Class	Witness Procedure	Witness Cost W
Nominal-tag	Single tag read	$O(1)$
Attribute-only	Query a distinguishing family of minimum size	$\Omega(d)$
Canonical axis core	Query any matroid basis of the core	exact basis cardinality

TABLE I
WITNESS COST FOR CLASS IDENTITY BY OBSERVER CLASS.

V. DISTORTION CURRENCY AND ENTROPY FLOORS

A. Distortion as a Currency

a) Currency comparison.: The counting and entropy converses above show that auxiliary bits, queries, and accepted distortion are alternative currencies for the same residual ambiguity. This section makes one exchange explicit: how many bits must be paid to drive distortion to zero on a collision fiber, and what floor remains when those bits are withheld.

The general law is fiberwise. For arbitrary finite sources, the optimal recoverable mass decomposes exactly across representation fibers. The clean closed form arises in the uniform single-block corner, where the collision geometry alone determines the full budget-distortion tradeoff.

B. Finite Fiberwise Decomposition

The adaptive and fixed-length theorems above quantify the residual ambiguity in each representation fiber separately. We now state the structural law that governs the global rate-distortion problem: for any source distribution and any finite set of fibers, the globally optimal recoverable mass decomposes exactly into fiberwise contributions, and this optimum is attained by independent per-fiber selection.

This is the main structural theorem of the rate-distortion arc. It applies to arbitrary finite sources and arbitrary observation maps, not only to uniform sources on single collision blocks.

Definition V.1 (Fiberwise feasible subset). Fix a uniform per-fiber tag alphabet of size T . A subset $S \subseteq \mathcal{C}$ is *fiber-budget feasible* if every representation fiber contains at most T elements

of S :

$$|S \cap \pi^{-1}(u)| \leq T \quad \text{for all } u.$$

Definition V.2 (Fiberwise optimum). For each representation value u , define the *fiberwise top mass*

$$M^*(u, T) := \max \left\{ \sum_{c \in S} \mu(c) : S \subseteq \pi^{-1}(u), |S| \leq T \right\}.$$

This is the maximum source mass recoverable inside fiber u using at most T tag values.

Theorem V.3 (Fiberwise decomposition). *Let μ be any finite source distribution and let T be a uniform per-fiber tag budget. Then the global optimum under the fiber-budget constraint is the sum of the fiberwise optima:*

$$M_{\text{global}}^*(T) = \sum_u M^*(u, T).$$

Moreover, any feasible subset S satisfies

$$\mu(S) \leq \sum_u M^*(u, T).$$

(L: [FRD1](#))

Proof. Any feasible subset decomposes into its fiber slices, each bounded by its fiberwise optimum. Summing over fibers gives the upper bound. Conversely, choosing a mass-maximizing subset of size at most T independently in each fiber yields a globally feasible subset attaining equality. ■

Corollary V.4 (Fiberwise attainment). *There exists a globally feasible subset S^* that attains the decomposition bound:*

$$\mu(S^*) = \sum_u M^*(u, T).$$

(L: [FRD2](#))

Proof. Take $S^* = \bigcup_u S_u^*$ where S_u^* is a maximizing subset in fiber u . The union is feasible because each fiber contributes at most T elements, and the mass is the sum of the fiberwise

maxima. ■

Remark V.5 (The decomposition is the general law). The decomposition theorem applies to any finite source distribution, including uniform, Zipf, power-law, and arbitrary. The uniform single-block formula derived below is the special case where (i) all source mass lies in one fiber and (ii) the mass is uniformly distributed. Neither assumption is required for the decomposition itself. In this sense, the decomposition is the general zero-error rate-distortion law for the fixed-representation setting; the closed-form curve is its cleanest instantiation.

Remark V.6 (Relation to classical source coding). The decomposition is analogous to coding independent sub-sources separately: each fiber acts like an independent sub-problem, and the global optimum is the sum of the sub-problem optima. The sub-sources are induced by the representation partition rather than independent a priori.

C. The Uniform Single-Block Corner

The fiberwise decomposition theorem reduces the general problem to independent per-fiber optimization. When the source is uniform on a single collision block, this optimization collapses to a simple closed form. The following theorem is therefore a corollary of the structural law above, specialized to the uniform corner.

Theorem V.7 (Semantic identity rate-distortion theorem). *Let $G \subseteq \mathcal{C}$ be a collision block of size a , so all classes in G have the same representation profile. Assume the source class C is uniform on G . Then the optimal distortion under a fixed-length auxiliary budget of L bits is exactly*

$$D^*(L) = \max\left(0, 1 - \frac{2^L}{a}\right).$$

Equivalently, every fixed-length scheme with auxiliary budget L bits satisfies

$$\Pr[\hat{C} = C] \leq \frac{2^L}{a}, \quad D = \Pr[\hat{C} \neq C] \geq 1 - \frac{2^L}{a}.$$

If $2^L \geq a$, then zero distortion is achievable; if $2^L < a$, then the lower bound is tight.

(L: [GPH30](#))

Proof. Fix a realization of all internal randomness used by the scheme. Because every class in G has the same representation profile, the decoder's side information is the same on all classes in G ; on that block, the decoder can therefore distinguish classes only through the auxiliary outcome. With L bits there are at most 2^L outcomes, and for each such outcome the decoder can be correct on at most one class in G . Hence, for the fixed randomness, the scheme is correct on at most 2^L of the a classes in G , so its success probability under the uniform distribution on G is at most $2^L/a$. Averaging over the scheme's randomness preserves the same bound. The distortion lower bound is the complement of the success bound.

If $2^L \geq a$, assign distinct auxiliary outcomes to the classes in G and decode injectively; this gives zero distortion. If $2^L < a$, choose 2^L classes in G , assign them distinct auxiliary outcomes, and map all remaining classes to arbitrary auxiliary outcomes. Then the decoder is correct on exactly 2^L of the a classes, so the success probability is $2^L/a$ and the distortion is exactly $1 - 2^L/a$. This matches the converse. ■

Corollary V.8 (Zero-error threshold). *Under the hypotheses of Theorem V.7, zero distortion is achievable if and only if*

$$L \geq \log_2 a.$$

Proof. By Theorem V.7, zero distortion is achievable exactly when $D^*(L) = 0$, equivalently when $2^L \geq a$, which is the same as $L \geq \log_2 a$. ■

Corollary V.9 (Zero-budget error floor). *If $L = 0$ and the source is uniform on a collision block of size a , then any scheme relying only on the representation satisfies*

$$D \geq 1 - \frac{1}{a}.$$

In particular, if the source is uniform on a worst collision block with $a = A_\pi$, then

$$D \geq 1 - \frac{1}{A_\pi}.$$

(L: GPH30)

Proof. Set $L = 0$ in Theorem V.7. Since $2^0 = 1$, the distortion lower bound becomes $1 - 1/a$. Choosing a worst collision block gives the second statement. ■

Remark V.10 (The cost of omission). This corollary quantifies the price of relying exclusively on semantic attributes. If one chooses not to store the auxiliary identifier required by the fixed-length coding law, the saved $\lceil \log_2 A_\pi \rceil$ bits are exchanged for a nonzero error floor. On a collision fiber of size 2, omitting the one-bit tag forces an error rate of at least 50%. On a collision fiber of size 100, omitting the seven-bit tag forces an error rate of at least 99%. One pays for identity either in stored bits or in distortion. There is no third option in which identity is both free and exact.

Remark V.11 (Why this matters). The distortion theorem is not only a converse to the zero-error law. It identifies the operational cost of omission in semantics-aware compression systems. If a system declines to store the identity bits needed to separate collided fibers, then the penalty is not abstract suboptimality but a concrete minimum error rate. The important feature is the steepness of the curve: because distortion scales as $1 - 2^L/a$, small deficits in the coding budget can produce large losses in identity accuracy. This is the exact storage-versus-distortion tradeoff induced by semantic abstraction.

Remark V.12 (Steepness). The steepness is easy to see numerically. On a collision block of size 100, zero distortion requires at least $\lceil \log_2 100 \rceil = 7$ bits. With only 6 bits the distortion is at least 36%, and with only 5 bits it is at least 68%. Small shortfalls below the exact-recovery threshold can therefore make identity accuracy unusable in practice.

Remark V.13 (Scope of the exact curve). The exact formula above is stated for the uniform source on a single collision block. That restriction is intentional. It isolates a deterministic corner in which collision geometry alone determines the full tradeoff. For non-uniform sources on the same block, or for mixtures across multiple fibers, the optimal distortion curve depends on the source law as well as the fiber sizes; in those settings one expects a different shape, closer in spirit to entropy-weighted side-information bounds than to the uniform worst-block law treated here. The point of the present theorem is exact isolation of the geometry-only effect.

D. Finite Entropy Converses

The counting-based converses above are combinatorial: they bound distortion in terms of fiber sizes and tag budgets. A complementary family of converses connects distortion to information-theoretic quantities, placing the semantic identity problem inside classical Shannon theory.

The following theorems are mechanized in the RDC series, from [RDC1](#) through [RDC9](#). They apply to any finite source, any observation map, and any tag/decoder pair.

Theorem V.14 (Fano-style finite converse). *Let μ be a finite source on K classes, let (O, T) be the observation and tag alphabet sizes, and let D be the distortion (error probability) of any deterministic decoder. Then*

$$H_\mu(C) \leq h_b(D) + (1 - D) \log(O \cdot T) + D \log(K - 1),$$

where $H_\mu(C)$ is the source entropy and h_b is binary entropy.

(L: [RDC1](#))

Proof. Partition the source into success and failure sets. The success set has cardinality at most $O \cdot T$ (injection into observation/tag pairs), and the failure set has cardinality at most $K - 1$. Applying the entropy bound for each partition and combining via the chain rule yields the stated inequality. The mechanization follows the standard Fano argument adapted to the finite budgeted setting. ■

Corollary V.15 (Budget lower bound from distortion). *Under the hypotheses of Theorem [V.14](#), if $D < 1$ then*

$$H_\mu(C) - h_b(D) - D \log(K - 1) \leq \log(O \cdot T).$$

(L: [RDC6-7](#))

Proof. Rearrange Theorem [V.14](#). The left-hand side is the “entropy gap” that must be absorbed by the observation/tag budget. ■

Theorem V.16 (Observation-only min-entropy bound). *For observation-only schemes (no tags, $T = 1$), any decoder with error probability at most $\varepsilon < 1$ satisfies*

$$H_\infty(\mu) \leq -\log(1 - \varepsilon),$$

where $H_\infty(\mu)$ is the min-entropy of the source.

(L: [RDC9](#))

Proof. The success probability is bounded by the maximum atom mass times the budget (unity, for observation-only). Rearranging and taking logarithms gives the min-entropy bound. ■

Remark V.17 (Counting versus entropy converses). The counting converse $2^L \geq a$ is sharp in the zero-error regime but silent about source structure. The entropy converses fill this gap. Distortion is governed by how much entropy the observation/tag pair can carry. For high-entropy sources with large collision blocks, the entropy bound may be tighter than the counting bound; for low-entropy sources with concentrated mass, the counting bound dominates. Together, the two converse families provide complementary views of the same fiber geometry, one combinatorial and one information-theoretic.

Theorem [V.14](#) is a finite, deterministic, zero-error variant of the classical Fano inequality. The binary entropy term $h_b(D)$ captures uncertainty about decoder success; the $\log(O \cdot T)$ term is the rate of the observation/tag channel; the $\log(K - 1)$ is the standard Fano confusion penalty.

E. Worked Example

Return to the running example with profiles

$$A = 000, \quad B = 000, \quad C = 100, \quad D = 110, \quad E = 111.$$

The fixed-length theorem gives $A_\pi = 2$, so one extra bit is necessary and sufficient for exact recovery. The adaptive theorem says more: the fiber over profile 000 requires one bit, while the singleton fibers over 100, 110, and 111 require zero bits. Thus the expected auxiliary rate is exactly the probability of landing in the collided fiber.

This is the representation-level interpretation the paper emphasizes. The representation already resolves three of the four fibers completely; the only irreducible metadata cost is the one residual collision between A and B .

The fiberwise decomposition theorem gives the same result structurally: the global optimum is the sum of the fiberwise optima, with the collided fiber contributing its top- T mass and the singleton fibers contributing their full mass. The uniform single-block formula is the special case where all source mass lies in one fiber.

The next section turns this coding object into a representation-sufficiency theory: once the residual ambiguity of each fiber is explicit, one can ask exactly which downstream tasks, helper views, and modular factors eliminate it and which do not.

VI. SYSTEMS CONSEQUENCES: WHY SYMBOLS ARE NECESSARY

The formal and coding results lead to a practical systems corollary: if a representation π is non-injective, then no purely semantic pipeline can guarantee zero-error identity recovery. The identity debt must be paid. In deployed systems the canonical payment mechanisms are symbolic handles, such as unique identifiers, pointers, retrieval keys, primary keys, and registry entries.

This is not an engineering preference but a coding consequence. If exact identity is required in an open world, the system must either use an injective encoder, store or transmit explicit identity metadata of length at least $\log_2 A_\pi$ in the worst case, or accept a nonzero distortion floor. The same logic applies on the query side: one cannot escape the cost by redescribing the same observables redundantly, because the query theorem reduces any complete derivability-aware observable family to its orthogonal minimal core.

Example. In a retrieval system with 100 documents sharing the same embedding, omitting the $\lceil \log_2 100 \rceil = 7$ identifier bits leaves a distortion floor of $1 - 1/100 = 99\%$ on that collision fiber. Even one bit below the exact-recovery threshold is costly: with only 6 bits the distortion is still at least 36%.

A. *Learned Representations and Formal Instantiations*

The abstract coding framework applies to any representation map. In the learned-representation setting, semantic information is what the encoder exposes, while identity information is the exact class label or entity name that may still need to be carried explicitly. Zero-error recovery from semantic information alone is possible exactly when the semantic representation is injective on identity classes. When it is not, the collision fibers quantify the irreducible identity residue that must be restored by explicit metadata.

This transfers directly to neurosymbolic systems. A neural encoder supplies the semantic side by producing a learned representation; a symbolic layer attempts to recover or reason about exact class-level distinctions. The fiber geometry says precisely when this is possible without extra scaffolding and when it is not. Symbolic reasoning cannot recover distinctions already collapsed by the encoder except by receiving additional identifiers, additional observables, or additional disclosures.

a) What this means for ML systems.: The coding laws have immediate consequences for learned-representation systems.

- **Concept bottleneck models.** If the intermediate concept representation has collisions, then no downstream classifier can distinguish the collapsed concepts without extra identity information. Exact concept-level decisions therefore require an added disambiguation channel, retrieval key, or explicit symbolic concept identifier.
- **Retrieval-augmented systems.** The retrieval index pays the identity-bit budget. If two artifacts have the same embedding, the system must preserve distinguishing metadata elsewhere, such as document identifiers, source pointers, or exact lookup keys. Otherwise exact retrieval is impossible on the collided fiber.
- **Pure representation schemes.** Systems that attempt to avoid any nominal layer do not escape the theorem. They may still be useful, but they cannot guarantee zero-error open-world identity when the representation is non-injective.
- **Open-world deployments.** Any system that must identify entities exactly in an open world must either use an injective encoder or carry explicit identity metadata. There is no third

option.

b) Attribute-only versus tag-augmented design.: Whenever $A_\pi > 1$, the coding laws imply a sharp design dichotomy. In an attribute-only strategy, the system stores or transmits only the semantic representation $\pi(v)$. This pays only for semantic description, but it inherits the distortion floor forced by the collision fibers; in the zero-budget case, the worst collided block incurs error at least $1 - 1/A_\pi$ by Corollary V.9. In a tag-augmented strategy, the system stores $\pi(v)$ together with auxiliary identity information. This increases storage by the exact identity budget, up to $\lceil \log_2 A_\pi \rceil$ bits in the worst case, but restores zero-error identity recovery.

This is why symbolic identifiers are ubiquitous in high-reliability systems. Primary keys, memory addresses, retrieval identifiers, and similar handles are the mechanisms by which systems avoid the distortion floor mandated by the coding laws. If the application requires exact retrieval, verification, provenance tracking, or entity-level consistency, the attribute-only strategy is infeasible and the tag-augmented strategy is mandatory.

c) Formal instantiation.: The paper also develops a concrete formal identity domain in Lean. In that model, entities carry an explicit attribute family $\mathcal{I}$, primitive observables are the associated binary maps q_I , and the representation is the induced profile map $\pi(v) = (q_I(v))_{I \in \mathcal{I}}$. In this setting, representation-computable properties reduce to attribute-computable ones, and the generic coding laws specialize to a machine-checked formal instance.

The formal theorems establish that equal observable profiles cannot distinguish identity (ACS8), that provenance is not recoverable from the profile alone when collisions exist (ACS9), and that nominal identity cannot be embedded back into the semantic layer (EMB1). Readers interested only in the information-theoretic content may treat this development as a rigorous formal instantiation rather than as a prerequisite for the main theorem arc.

d) Instantiated neurosymbolic linking theorems.: Within that formal identity domain, the generic helper-view theorems specialize by setting the task to identity $Y = C$. The mechanization (NSL1 through NSL6) proves the following instantiated claims.

- If the symbolic handle σ is injective, the neurosymbolic link (π, σ) achieves zero-error identity recovery (NSL1). This is the identity-task instantiation of the injectivity criterion

GPH25.

- If π has collisions, semantic representation alone cannot achieve zero-error identity (NSL2), and no decoder can recover the collapsed identities without identity-resolving side information (NSL3).
- The exact characterization is: (π, σ) recovers identity if and only if σ distinguishes all entities within each semantic fiber (NSL4). An injective handle therefore suffices (NSL5). This is the identity-task instantiation of the helper-view theorem GPH40.
- The pair (π, σ) is the canonical helper view for open-world identity systems (NSL6).

These instantiated theorems connect the generic information-theoretic helper-view framework to a concrete formal identity domain. Together with ACS8, ACS9, and EMB1, they show how the auxiliary identity channel pays for the ambiguity that the semantic layer leaves unresolved.

B. Task Sufficiency, Helper Views, and Factorization

The neurosymbolic reading is one specialization of a broader theorem family. The same fiber geometry characterizes when a representation is sufficient for an arbitrary downstream task, when helper views genuinely help, and when factorized modules contribute real exact information.

Definition VI.1 (Task-fiber ambiguity). Let $Y = f(C)$ be a downstream task label induced by the class. For a fixed representation π , define

$$A_{\pi \rightarrow Y} := \max_u |\{f(c) : \pi(c) = u\}|.$$

This is the maximum number of task labels that remain unresolved inside one representation fiber.

Theorem VI.2 (Task sufficiency criterion). *The representation alone supports zero-error recovery of Y if and only if $A_{\pi \rightarrow Y} \leq 1$.*

(L: GPH38)

Proof. Zero-error recovery of Y from U alone is possible exactly when every representation fiber carries a single task label. That condition is equivalent to the statement that the largest

task-fiber ambiguity is at most one. ■

Corollary VI.3 (Task-level coarsening monotonicity). *If $\pi_{\text{coarse}} = g \circ \pi_{\text{fine}}$, then*

$$A_{\pi_{\text{fine}} \rightarrow Y} \leq A_{\pi_{\text{coarse}} \rightarrow Y}.$$

In particular, coarsening a representation cannot improve exact downstream task sufficiency.

(L: [GPH39](#))

Proof. Every coarse fiber is a union of fine fibers, so collapsing from π_{fine} to π_{coarse} cannot reduce the number of task labels realizable inside the worst fiber. ■

Theorem VI.4 (Helper-view sufficiency). *Let $S = \sigma(C)$ be an additional deterministic view available to the decoder. Then exact recovery of Y from (U, S) is possible if and only if each joint fiber of (π, σ) carries at most one task label.*

(L: [GPH40](#))

Proof. The pair (U, S) is just a refined deterministic representation. Exact recovery is therefore possible exactly when the downstream task is constant on the fibers of that refined representation.

■

Corollary VI.5 (Helper-view monotonicity). *Adding a deterministic helper view cannot worsen task ambiguity. In particular, the task-fiber ambiguity under (U, S) is at most the task-fiber ambiguity under U alone.*

(L: [GPH41](#))

Proof. Passing from U to (U, S) refines the representation fibers, so the largest number of task labels per fiber can only decrease. ■

Corollary VI.6 (Factorized product law). *In a genuinely factorized product setting, where the class space decomposes as a product and the representation decomposes coordinatewise, the joint*

fiber is the product of the component fibers and the worst-case residual ambiguity multiplies exactly across modules.

(L: [GPH42-43](#))

Proof. For coordinatewise product representations, each joint fiber is exactly the product of the corresponding component fibers, so their cardinalities multiply. Taking the largest such product yields multiplicativity of the worst-case residual ambiguity. ■

For learned representations, estimating quantities such as A_π , $A_{\pi \rightarrow Y}$, or the adaptive fiber profile is itself a representation-analysis problem rather than part of the theorem. In a finite explicit registry or index this is just fiber counting; for a large learned embedding one would need either exact collision enumeration after discretization or provable upper and lower bounds on the induced fibers. The point of the present results is to identify what those quantities mean once they are known.

Read concretely, the task-sufficiency theorem is the exact-recovery analogue of a sufficient-statistic condition for the downstream task Y : the representation is sufficient precisely when every fiber carries one task label. The helper-view and factorized laws then say when concept bottlenecks, retrieval-augmented pipelines, multimodal encoders, or modular neural architectures contribute genuinely new exact information rather than repackaging ambiguity already present in the base representation.

In the finite explicit setting, one-symbol zero-error feasibility is equivalent to injectivity and refinement cannot increase collision multiplicity ([GPH27](#), [GPH28](#)). The same setting also yields a disclosure separation between nominal access and tag-free identity resolution ([PRIV3](#)).

C. One Additional Semantic-Identification Example

Consider a model registry in which artifacts are indexed by a semantic representation π built from declared capabilities or structural signatures. If two distinct artifacts have the same representation value, then the representation alone cannot support exact registry identification. The fixed-length theorem says that the worst-case auxiliary identifier length is determined by the

largest collision bucket. The adaptive theorem says that rare collision buckets contribute little to expected metadata cost, while large or frequent buckets dominate it.

This example is close to the intended task-aware compression reading: the representation is already useful for search and organization, but exact identification may still require additional stored metadata. The theorem identifies the exact price of that residual ambiguity.

VII. ROBUSTNESS UNDER SYSTEM GROWTH

A. Secondary Robustness and Decidability Boundary

The main side-information theorems address whatever collision structure is present in the realized representation. A complementary question is whether that fiber geometry is stable under system evolution: can a representation that is currently collision-free become noninjective when the class universe grows? The next two results are secondary robustness statements quantifying that fragility.

Definition VII.1 (Finite realized world). Let $\mathcal{C}$ be the class universe. A *finite realized world* is a finite subset $W \subseteq \mathcal{C}$ of classes currently present in a deployed system snapshot. We call W *barrier-free* iff

$$\forall c_1, c_2 \in W, \pi_{\mathcal{C}}(c_1) = \pi_{\mathcal{C}}(c_2) \Rightarrow c_1 = c_2.$$

(L: [ROB1](#))

Definition VII.2 (Open-world extension). An *open-world extension* is a super-world $W' \supseteq W$ obtained by adding classes while keeping the observation family Φ fixed.

(L: [ROB1-2](#))

Theorem VII.3 (Barrier-freedom is not extension-stable). *In the open-world class model used here, it is not true that barrier-freedom is preserved under all extensions:*

$$\neg \left(\forall W \subseteq W', \text{BarrierFree}(W) \Rightarrow \text{BarrierFree}(W') \right).$$

(L: [ROB1-2](#))

Proof. Take the empty world $W_0 = \emptyset$, which is barrier-free. In this model, two concrete classes are shape-equivalent (same observable profile) but nominally distinct, so adding them yields an extension $W_1 \supseteq W_0$ that is not barrier-free. Therefore barrier-freeness is not extension-stable. ■

Remark VII.4 (Safety interpretation). The theorem formalizes a basic fact about open-world systems: present collision-freeness does not by itself yield a stable future guarantee. Persistent zero-error claims therefore need an explicit growth model rather than a one-time snapshot argument.

The running example offers the same intuition in finite form: the realized world $\{A, C, D, E\}$ is collision-free, but adding class B immediately restores the collision block $\{A, B\}$.

For compression system design, the point is direct: a representation that currently needs no auxiliary identification bits can require a positive residual coding budget after extension if new classes enter existing fibers. Thus open-world growth can change the optimal exact side-information rate even when nothing about the deployed decoder changes.

Definition VII.5 (Function-level barrier predicate). For a partial observer generator $f : \mathbb{N} \rightharpoonup \mathbb{N}$, define

$$\text{HasBarrier}(f) :\Leftrightarrow \exists x \neq y, \exists z, z \in f(x) \wedge z \in f(y).$$

Define $\text{BarrierFree}(f) :\Leftrightarrow \neg \text{HasBarrier}(f)$.

(L: [UND1-2](#))

Theorem VII.6 (Rice-style non-computability of barrier certification). *There is no computable predicate on program codes deciding whether the generated observer has a barrier:*

$$\neg \text{ComputablePred}(c \mapsto \text{HasBarrier}(\text{eval}(c))).$$

Equivalently, barrier-freeness certification is also non-computable.

(L: [UND1-2](#))

Proof. Assume there is a computable certifier for $P(c) := \text{HasBarrier}(\text{eval}(c))$. The property is non-trivial: some generators produce collisions and some do not. Rice’s theorem therefore implies that if membership in the corresponding semantic class were computable from code, every partial recursive function would have to satisfy it, contradiction. The barrier-free statement is equivalent by negation. ■

Remark VII.7 (Certification interpretation). The Rice-style theorem is the algorithmic counterpart to extension instability. Even when a current deployment appears collision-free, there is no general computable procedure that certifies this property for arbitrary future-generating code. Formal guarantees therefore require either a restricted extension model or explicit identity-carrying information that remains valid under growth.

Corollary VII.8 (Certification consequence for persistent $D = 0$ claims). *For unrestricted open-world generators, one cannot computably certify that attribute-only identification remains barrier-free under all future extensions. Consequently, persistent $D = 0$ guarantees cannot rely on “currently collision-free” attribute structure alone.*

(L: [ROB1-2](#), [UND1-2](#))

VIII. CONCLUSION

This paper identifies the structural object governing zero-error identity recovery from fixed semantic representations: the collision fiber geometry. From this single object flow exact laws for multiple operational modalities.

a) The unified view.: The same fiber geometry that forces $\log_2 A_\pi$ auxiliary bits also determines the query cost of operational disambiguation, the distortion floor $1 - 2^L/a$, and the stability conditions under growth. These are not separate phenomena but different currencies for the same residual ambiguity. If the representation is injective, identity is free. If the representation is non-injective, the missing information must be paid somewhere: in auxiliary bits, in query effort, in helper views, or in distortion. No amount of downstream processing can remove this cost without paying in one of these currencies.

On the query side, the paper proves something stronger than a restricted convenience theorem. Complete derivability-aware query families reduce to orthogonal semantically minimal cores. On that canonical core, exchange holds and matroid basis cardinality becomes the exact query invariant. Non-orthogonal query families are therefore redundant overpresentations, not a rival foundational regime.

The broader payoff is representation sufficiency. The same ambiguity-fiber geometry governs when a representation is sufficient for exact downstream tasks, when helper views genuinely help, and when modular or factorized latents add exact information rather than architectural redundancy. In the learned-representation reading, the interpretation is direct: injective representations support exact downstream identification with no extra metadata, while non-injective representations impose an irreducible side-information cost and, if that cost is left unpaid, a correspondingly irreducible distortion floor.

b) The neurosymbolic necessity theorem.: The central systems-level contribution is a formal justification for symbolic mechanisms in semantic systems. The coding laws prove that semantic abstraction imposes an exact identity cost: a non-injective representation requires auxiliary description to resolve its collision fibers, with worst-case cost $\log_2 A_\pi$ bits and fiberwise cost $\lceil \log_2 |\pi^{-1}(u)| \rceil$. Systems that require both semantic generalization and exact identity must therefore implement a mechanism to pay this cost.

Symbolic handles, identifiers, pointers, keys, and index entries are standard computational instantiations of that payment. Consequently, the integration of symbolic identity layers into neurosymbolic, retrieval, and other semantics-aware systems follows directly from the coding laws. The steepness of the distortion curve explains the practical importance: because distortion scales as $1 - 2^L/a$, even a small shortfall below the identity threshold can make exact identity unusable in practice. In the instantiated formal domain, the pair consisting of a neural encoder and an injective symbolic handle achieves zero-error identity recovery if and only if the handle distinguishes entities within each semantic fiber ([NSL1](#) through [NSL6](#)).

c) Mechanization.: The main theorem chain is machine-checked in Lean 4. The mechanization confirms that the fiber-geometry framework, the canonical query-core reduction, and

the neurosymbolic linking results are not informal analogies but exact formal statements.

Acknowledgment: AI-use Disclosure

Generative AI tools (including Codex, Claude Code, Augment, Kilo, and OpenCode) were used throughout this manuscript, across all sections (Abstract, Introduction, theoretical development, proof sketches, applications, conclusion, and appendix) and across all stages from initial drafting to final revision. The tools were used for boilerplate generation, prose and notation refinement, LaTeX and structure cleanup, translation of informal proof ideas into candidate formal artifacts (Lean and LaTeX), and repeated adversarial reviewer-style critique passes to identify blind spots and clarity gaps.

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion or exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review; Lean was used as an integrity gate for responsible AI-assisted research. The author is solely responsible for all statements, citations, and conclusions.

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Lean Handle Index.

ID	Handle	ID	Handle
ACS1	adversary_forces_n_minus_l_queries AbstractClassSystem/Extended.lean	ACS7	nominal_localization_constant_semantic AbstractClassSystem/Extended.lean
ACS8	shape_cannot_distinguish AbstractClassSystem/Core.lean	ACS9	shape_provenance_impossible AbstractClassSystem/Core.lean
EMB1	semantic_non_embeddability AbstractClassSystem/Typing.lean	FRD1	ObserverModel.feasible_subsetMass_le_optimalFeasibleMass Paper1IT/FiberRateDistortion.lean
FRD2	ObserverModel.optimalSubset_attains_optimalFeasibleMass Paper1IT/FiberRateDistortion.lean	GPH13	Ssot.GraphEntropy.tagFeasible_iff_maxFiberCard_le
GPH16	Ssot.GraphEntropy.block_tagFeasible_iff_pow_maxFiberCard_le Paper1IT/GraphEntropyAsymptotic.lean	GPH17	Ssot.GraphEntropy.maxFiberCard_blockObserve_eq_pow Paper1IT/GraphEntropyAsymptotic.lean
GPH18	Ssot.GraphEntropy.blockTagRateBits_eq_mul_oneShot Paper1IT/GraphEntropyAsymptotic.lean	GPH19	Ssot.GraphEntropy.blockTagRateBitsPerCoordinate_eq_oneShot Paper1IT/GraphEntropyAsymptotic.lean
GPH20	Ssot.GraphEntropy.minBlockFeasibleAlphabet_eq_pow Paper1IT/GraphEntropyAsymptotic.lean	GPH24	Ssot.GraphEntropy.feasibleAtAlphabetBase_iff Paper1IT/GraphEntropyAsymptotic.lean
GPH25	Ssot.GraphEntropy.maxFiberCard_le_one_of_injective	GPH27	Ssot.GraphEntropy.tagFeasible_one_iff_injective
GPH28	Ssot.GraphEntropy.maxFiberCard_mono_of_factors_through	GPH30	Ssot.GraphEntropy.uniformFiberErrorRate_ge_one_sub_tag_ratio
GPH31	Ssot.GraphEntropy.inclusionMinimalCoordinateDistinguishing_not_equicardinal	GPH32	Ssot.GraphEntropy.optimalFiberBitLength_feasible
GPH33	Ssot.GraphEntropy.optimalExpectedAdaptiveBitLength_le	GPH34	Ssot.GraphEntropy.conditionalEntropyGiven_le_log2_mul_expectedAdaptiveBitLength
GPH35	Ssot.GraphEntropy.exists_conditionalCodes_expectedLength_le_entropy_bits_plus_one	GPH37	Ssot.GraphEntropy.optimalFiberBitLength_mono_of_factors_through
GPH38	Ssot.GraphEntropy.taskRecoverable_iff_maxTaskFiberCard_le_one	GPH39	Ssot.GraphEntropy.maxTaskFiberCard_mono_of_factors_through
GPH40	Ssot.GraphEntropy.taskRecoverable_pair_iff	GPH41	Ssot.GraphEntropy.maxTaskFiberCard_pair_le_left
GPH42	Ssot.GraphEntropy.observeFiber_prod_card	GPH43	Ssot.GraphEntropy.maxFiberCard_prod
GPH44	Ssot.GraphEntropy.maxFiberCard_eq_finiteSup Paper1IT/ComputabilityQuantization.lean	GPH45	Ssot.GraphEntropy.maxFiberCard_computable_by_fiber_enumeration Paper1IT/ComputabilityQuantization.lean
GPH48	Ssot.GraphEntropy.quantization_lower_bound_fin Paper1IT/ComputabilityQuantization.lean	GPH50	Ssot.GraphEntropy.precision_requirement Paper1IT/ComputabilityQuantization.lean
GPH52	Ssot.GraphEntropy.fiber_is_clique	GPH53	Ssot.GraphEntropy.fiber_finset_is_clique
L1	matroid_basis_equicardinality axis_framework.lean	L4	l4_exchange_wrapper HandleAliases.lean
L5	l5_exchange_wrapper HandleAliases.lean	L6	nonorthogonal_complete_has_redundant_axis axis_framework.lean
L7	exists_semanticallyMinimal_subset axis_framework.lean	L8	exists_orthogonal_semanticallyMinimal_subset axis_framework.lean
LWDC1	LWDConverse.collision_block_requires_bits lwd_converse.lean	LWDC3	LWDConverse.maximal_barrier_requires_bits lwd_converse.lean
NSL1	AbstractClassSystem.Neuropsychological.neuropsychological_zero_error_identity AbstractClassSystem/Neuropsychological.lean	NSL2	AbstractClassSystem.Neuropsychological.shape_alone_not_zero_error AbstractClassSystem/Neuropsychological.lean
NSL3	AbstractClassSystem.Neuropsychological.neuropsychological_necessary_for_zero_error AbstractClassSystem/Neuropsychological.lean	NSL4	AbstractClassSystem.Neuropsychological.neuropsychological_identity_recovery AbstractClassSystem/Neuropsychological.lean
NSL5	AbstractClassSystem.Neuropsychological.neuropsychological_injective_implies_recoverable AbstractClassSystem/Neuropsychological.lean	NSL6	AbstractClassSystem.Neuropsychological.neuropsychological_is_canonical_helper_view AbstractClassSystem/Neuropsychological.lean
OM1	ObserverModel.pmfEntropy_source_eq_sourceEntropy Paper1IT/PMFEntropy.lean	PRIV3	AbstractClassSystem.identity_disclosure_separation AbstractClassSystem/Extended.lean
RDC1	ObserverModel.finiteRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean	RDC6	ObserverModel.finiteRateDistortionBound Paper1IT/FiniteRateDistortionBounds.lean
RDC7	ObserverModel.logBudgetLowerBoundFromError Paper1IT/FiniteRateDistortionBounds.lean	RDC9	ObserverModel.observationOnlyMinEntropyBound Paper1IT/RateDistortion.lean
ROB1	AbstractClassSystem.OpenWorld.extend_to_force_barrier AbstractClassSystem/Undecidability.lean	ROB2	AbstractClassSystem.OpenWorld.barrierFreedom_not_extension_stable AbstractClassSystem/Undecidability.lean
S1	Ssot.Paper1IT.poissonCellCollisionProb_eq_one_sub_exp_mul_one_add Paper1IT/GrowthCollisions.lean	UND1	AbstractClassSystem.OpenWorld.hasBarrier_not_computable AbstractClassSystem/Undecidability.lean
UND2	AbstractClassSystem.OpenWorld.barrierFree_not_computable AbstractClassSystem/Undecidability.lean	ZEC1	Ssot.GraphEntropy.zeroErrorConditionalEntropySandwich Paper1IT/ZeroErrorConditionalEntropy.lean